IBScovars: Shrinkage versus non−shrinkage

procedure	successful_repetitions	mean_interval_width	joint_benchmark_recovery_pct	mean_achieved_missing_pct
Non−shrinkage	10000	0.7433	99.96	30.0214
Shrinkage	10000	0.7523	99.97	30.0214

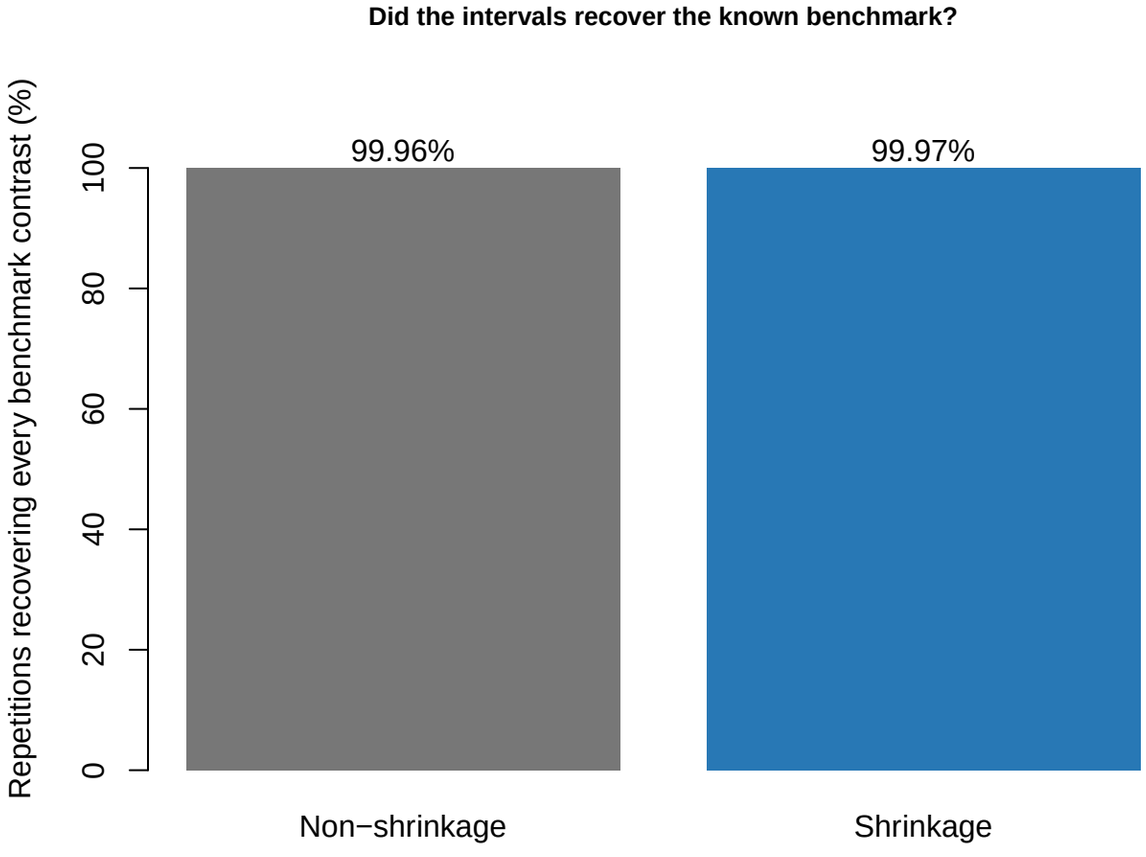
IBScovars: Paired comparison (Monte Carlo uncertainty)

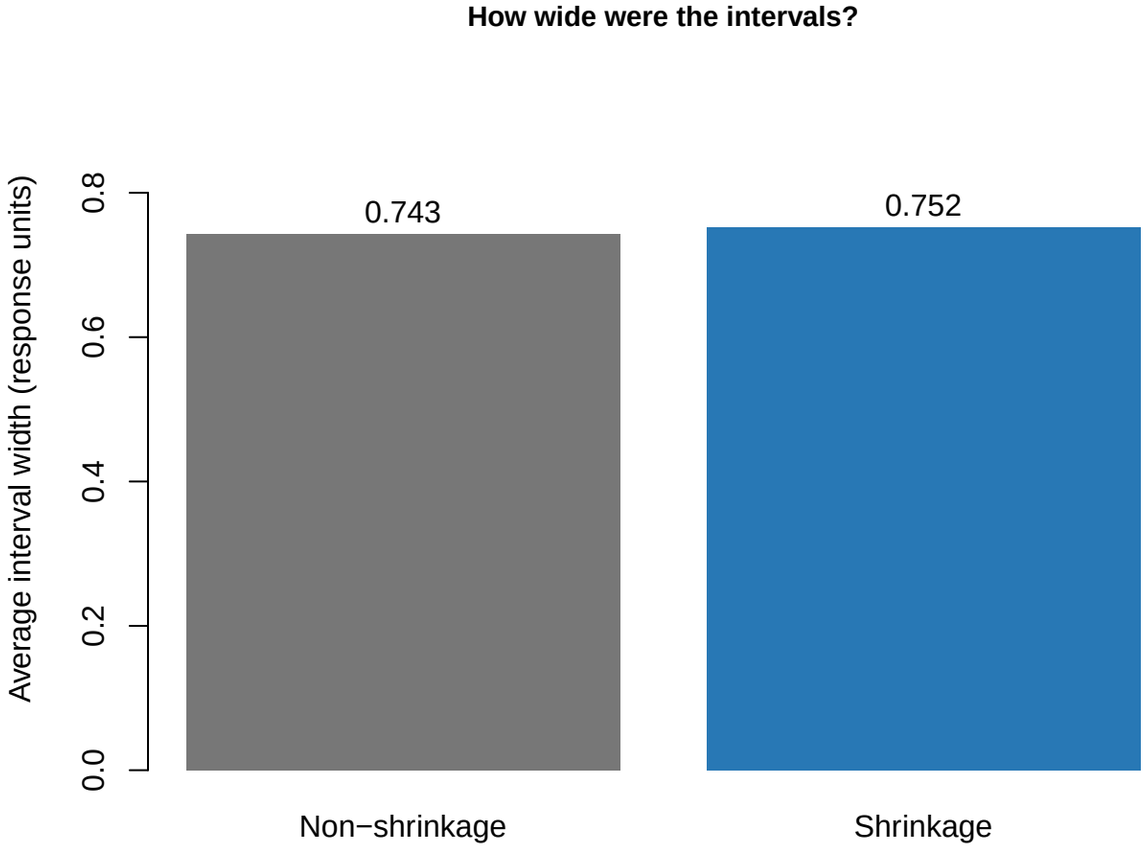
metric	estimate	monte_carlo_se	mc95_lower	mc95_upper
Width reduction: non−shrinkage minus shrinkage	−0.0089	0.000	−0.0090	−0.0089
Mean percentage width reduction	−1.2085	0.005	−1.2182	−1.1988
Repetitions in which shrinkage is narrower (%)	1.0300	0.101	0.8321	1.2279
Joint benchmark recovery difference: shrinkage minus non−shrinkage (percentage points)	0.0100	0.010	−0.0096	0.0296

IBScovars: Results by treatment versus control

variance_method	contrast	repetitions	mean_estimate	mean_width	benchmark_recovery_pct
Non−shrinkage	1 vs 0	10000	0.2844	0.7542	99.98
Non−shrinkage	2 vs 0	10000	0.2980	0.6953	100.00
Non−shrinkage	3 vs 0	10000	0.3515	0.7509	99.99
Non−shrinkage	4 vs 0	10000	0.3480	0.7729	99.99
Shrinkage	1 vs 0	10000	0.2844	0.7561	99.99
Shrinkage	2 vs 0	10000	0.2980	0.7209	100.00
Shrinkage	3 vs 0	10000	0.3515	0.7578	99.99
Shrinkage	4 vs 0	10000	0.3480	0.7742	99.99

IBScovars: 10000 repetitions; 40 regression imputations each; identical imputations for both procedures.





Shrinkage was 1.21% wider on average across repetitions.

New 30% expected MAR missingness in each repetition. Benchmark = original complete−data contrasts.
Benchmark recovery is not population−parameter coverage; higher recovery alone does not establish superiority.